

Course content and style

International Marketing Analytics



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[Office hours: book online \(Google Calendar\)](#)

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The journey begins





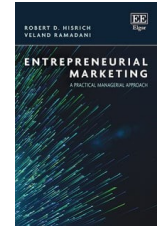
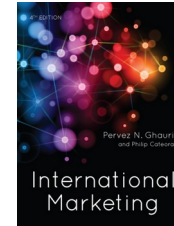
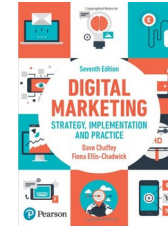
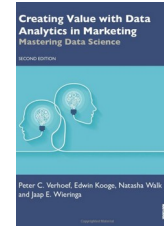
Personal and contact information

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Module Convenor - Dr Arne Floh



- Work Experience
 - International & Industry Background
 - Teaching Experience > 25 Years
- Teaching
 - International Marketing
 - International Business Projects
 - Marketing Research and Marketing Analytics
 - Global Branding
 - Supervision (BSc, MSc, PhD)
- Research
 - Dynamic Capabilities and Firm Performance
 - GenAI and Marketing
 - SMEs and Internationalization
- Private
 - Football
 - Electronic Gadgets
 - Austrian Wine





Marketing analytics turns marketing data into decisions. It measures what marketing achieved and shows where the next euro should go.



Key questions: how do price, place, product features and promotion drive performance measures such as sales, and how should a fixed budget be split across countries and channels?



Marketing mix modelling in plain words: a model that splits sales into what would have sold anyway and the extra sales each marketing activity brought in.

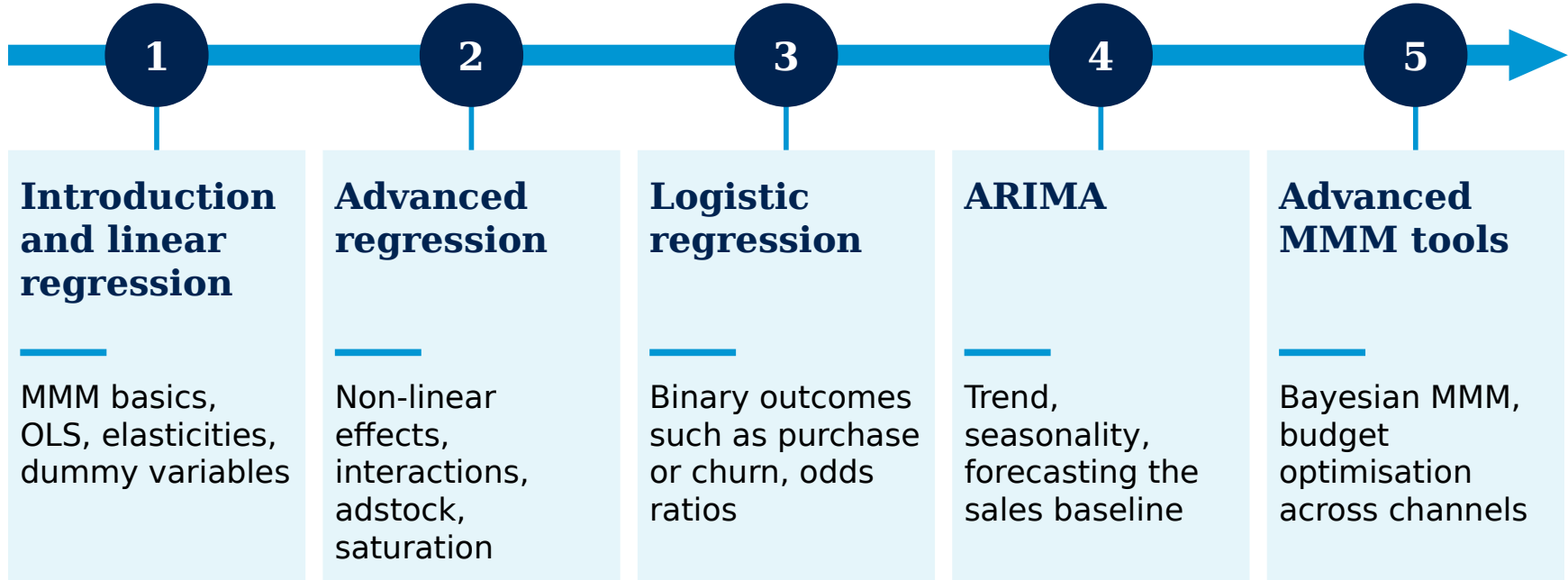


Tools: Python in Positron, GitHub for versioning and DataCamp courses; AI assistants do most of the typing, you check the results.



Case studies: Zalando and NEOH, plus a six-country teaching dataset with known answers for the labs.

Course roadmap: five sessions



Along the way: DataCamp certificates, coding exercises, guest speaker



At the end: written exam

Learning outcomes

On successful completion of this module, students will be able to:



Understand

- 1 **Explain the fundamentals of MMM**, its strengths and limits
- 2 **Describe real-world applications** of MMM in marketing



Apply

- 3 **Apply MMM in Python** to cross-sectional and longitudinal data
- 4 **Interpret results** and turn them into marketing decisions



Transfer

- 5 **Use AI tools** for analysis and reporting, and check the output
- 6 **Think critically** and work effectively in teams

Methods of teaching and learning

Every session

Concept lecture
60 min

Guided lab
90 min

Team exercise
60 min

Online quiz
not graded

Between sessions



- Read chapters and materials
- DataCamp courses and certificates
- Coding exercises
- Case study report in your group
- Q&A forum on Canvas



In the classroom



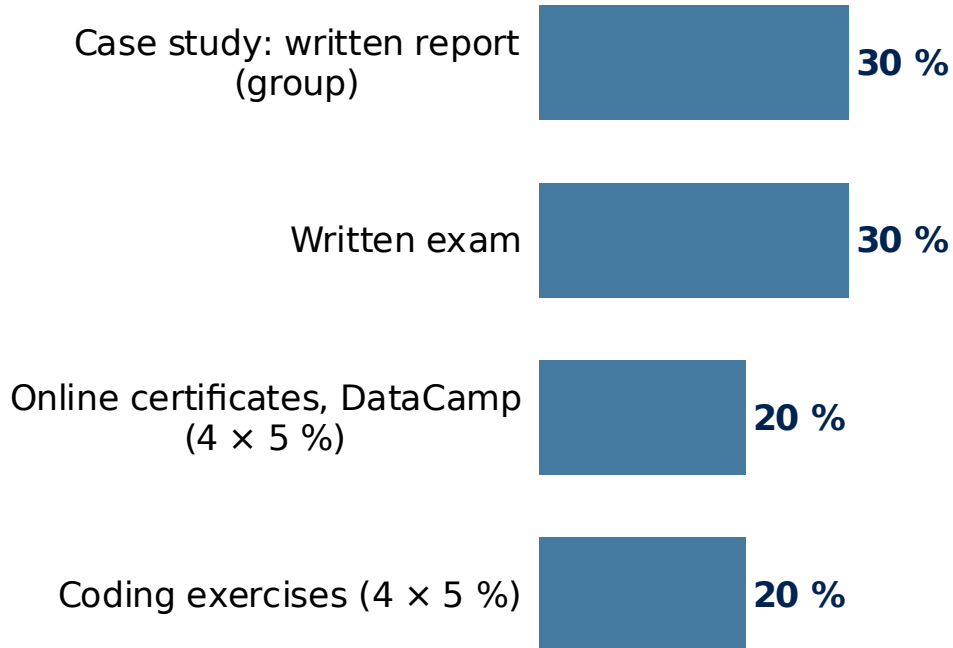
- Five in-person sessions
- Laptop with Python, Positron, GitHub
- AI-assisted coding, checked by you
- Guest speaker from industry



At the end: written exam, paper and pencil, multiple-choice questions.

Session	Topic	Methods
1	Introduction and linear regression	Course content and style; Python, Positron, GitHub and DataCamp; MMM fundamentals; OLS, log-log elasticities, dummy variables
2	Advanced regression	Non-linear effects, moderation and mediation, adstock and saturation, regression diagnostics
3	Logistic regression	Binary outcomes such as purchase or churn, odds ratios, model fit and classification
4	ARIMA	Trend, seasonality and autocorrelation; ARIMA and ARIMAX forecasts as the sales baseline
5	Advanced MMM tools	Bayesian MMM with PyMC-Marketing and Google Meridian; budget optimisation across channels and countries

Weight in the final grade



Rules



- **Individual:** certificates, coding exercises and the written exam
- **Group:** case study report, groups of 4 to 5; enrol on Canvas
- **Grades:** WU scale, 1 from 90 %, 2 from 80 %, 3 from 70 %, 4 from 60 %
- **Attendance:** at least 80 % of sessions
- **AI tools:** expected; you own every number you report

4

DataCamp courses. Python courses assigned on Canvas, each ending with a certificate.

5 %

Per certificate. Upload each certificate on Canvas by its deadline; 20 % of the grade in total.

45 min

Course environment. Install Python (via uv), Positron, Quarto and GitHub Desktop with the setup guide, then run the installation check.



Deadlines: the date for each certificate is on Canvas. Bring installation problems to Session 1.

Coding exercises and written exam

Coding exercises



- Four exercises during the course
- Apply the lab code to a similar task
- Submit your Python or Quarto file on Canvas
- AI assistants allowed; add a short prompt log

5 % per exercise, 20 % in total

Written exam



- At the end of the course
- Paper and pencil
- Multiple-choice questions
- Concepts and interpretation, not code

30 % of the final grade

Both are individual. The case study report (30 %) is the group component.

Case study: written report

4-5 students per
group; enrol on Canvas

30 % of the final
grade

1 written report per
group

What you submit

< > Quarto file
your analysis and text
in one reproducible
document

 **Rendered report**
the output of the
Quarto file; HTML
recommended

 **Data file**
the data you
analysed, so the
report can be rerun

The task: apply marketing mix modelling to a real marketing case, interpret the results and recommend a decision to management.

Support: online consulting hours (registration needed). Deadline: see Canvas.



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